

Implied Volatility Prediction using Machine Learning

Problem Overview: Predict implied volatility (IV) for options using historical option data using regression techniques.

Feature Engineering: Created non-linear features such as moneyness², log(moneyness), log(tau), and interaction terms. Time-based features like day-of-week and week-of-year were also included.

Market Regime Feature: Used date-wise average IV (date_iv_mean) with proper shifting to avoid leakage, capturing market conditions effectively.

Leakage Prevention: Applied time-based train-validation split ensuring no future information is used during training.

Model: XGBoost Regressor was used due to its strength in handling non-linear tabular data.

Performance: The model achieved an R² score of approximately **0.85**, indicating strong predictive performance without data leakage.

Conclusion: The approach successfully models implied volatility using robust feature engineering and proper validation strategy.